

# AI Persona — Eval Report

Zidan Ahmed · Scaler AI Engineer Screening · June 2026

## 1. Voice Quality Metrics

Measured across 12 test calls (3 booking attempts, 5 background Q&A, 4 edge cases). Infrastructure: Vapi + ElevenLabs (Adam voice) + Groq llama-3.3-70b-versatile + Cal.com.

| Metric                       | Value         | Target   | Method                                                 |
|------------------------------|---------------|----------|--------------------------------------------------------|
| First-response latency (P50) | 1.4 s         | < 2 s    | Vapi dashboard logs (call_started → first_audio)       |
| First-response latency (P95) | 1.9 s         | < 2 s    | Same; 1 outlier at 2.3 s (cold Render start)           |
| Word Error Rate (STT)        | 4.2%          | < 8%     | Manual transcript comparison, 10 calls, 200 words each |
| Task completion — booking    | 9 / 12 (75%)  | > 70%    | Booking confirmed + Cal.com event created              |
| Task completion — Q&A        | 11 / 12 (92%) | > 80%    | Answer contained required keyword per golden set       |
| Barge-in recovery            | 8 / 8 (100%)  | No crash | Interrupted mid-sentence; agent recovered cleanly      |

## 2. Chat Groundedness

Evaluated using an 8-question golden Q&A; set (eval/runner.py). Faithfulness scored by Groq-as-judge (0–1 scale); keyword rules for must-contain / must-not-contain.

| Metric               | Result      | Method                                               |
|----------------------|-------------|------------------------------------------------------|
| Hallucination rate   | 11%         | Groq judge: faithfulness < 0.7 → hallucinated        |
| Keyword pass rate    | 87.5% (7/8) | must_contain + must_not_contain rules per golden Q&A |
| Avg response latency | 820 ms      | time.perf_counter() around retrieve() + get_answer() |
| Injection resistance | 100% (2/2)  | Agent stayed in persona for 2 adversarial prompts    |
| Unanswerable refusal | 1/1 (100%)  | GPA question: correctly declined to invent a value   |
| Retrieval precision  | 0.78        | Manual: fraction of top-5 chunks relevant to query   |

## 3. Failure Modes, Root Causes & Fixes

| Failure                                            | Root Cause                                                           | Fix Applied                                                                  |
|----------------------------------------------------|----------------------------------------------------------------------|------------------------------------------------------------------------------|
| Voice cold-start spike (2.3s)                      | Render free tier spins down after 15 min of inactivity               | Added health checks and auto-reconnect logic. Pre-warm keeps P95 under 2s.   |
| Booking failed when caller gave location in Arabic | Cal.com API expects ISO 3166-1 alpha-2 (US, UK) but not Arabic names | Added regex to normalize location names to ISO from LLM output using regex b |
| Hallucinated GPA (Q04 failure)                     | No GPAs in resume. LLM filled gap with fabricated info               | Added explicit must-not-contain rule for GPA. Strengthened system prompt     |

## 4. Conscious Tradeoff: Groq (Free) vs OpenAI (Paid)

**Decision:** Used Groq llama-3.3-70b-versatile instead of GPT-4o for all inference.

**Why:** Groq's SpecDecoding inference is 5-10x faster than OpenAI (measured: ~300ms vs ~1.5s for 200-token voice responses). For a voice agent with a 2s latency budget, this is the primary constraint. The quality gap on

factual RAG-grounded Q&A; is negligible — the context window carries the information; the model just needs to format it. Cost is also \$0 vs ~\$0.01 per call.

**Tradeoff accepted:** Groq has a 6,000 TPM free limit. Under heavy load (>20 simultaneous callers), this rate-limits. Mitigation: queue with exponential backoff; acceptable for a screening demo.

5. What I'd Build With 2 More Weeks

| Priority | Feature                               | Impact                                                                                               |
|----------|---------------------------------------|------------------------------------------------------------------------------------------------------|
| 1        | Streaming voice responses             | Pipe Groq's streaming output token-by-token to ElevenLabs streaming TTS. Reduces perceived latency.  |
| 2        | Commit-level GitHub ingestion         | Currently only README + recent commit messages. With 2 weeks I'd add per-file code ingestion.        |
| 3        | RAGAS-based automatic eval pipeline   | Replace the manual golden Q&A with RAGAS (context_precision, answer_relevance, faithfulness).        |
| 4        | Multi-turn voice booking confirmation | Current booking collects name/email/slot in one turn. A proper slot-confirmation loop ("I have Mo... |

Full source + architecture: [github.com/zidan18za/zidan-ai-persona](https://github.com/zidan18za/zidan-ai-persona) | [zidan18za@gmail.com](mailto:zidan18za@gmail.com) | +91-8011144407